



Builder Basics — The Impossible Statue (Extension)

Topic: Place + Move, On and Off · **Course:** Builder Basics · **Type:** Extension Activity · **Time:** about 45 minutes

✖ This is a bonus challenge, not a weekly lesson. Try it once you are comfortable placing blocks and moving the agent.

1 · Read the World 🔍

Open the **Impossible Statue** world and read the markers:

- **Red blocks** — your **start** and **end** points.
- **Crying obsidian** — **build** here. Every dark spot is a block to place.
- **Glass** — a **trap**. Move through these spots but place **nothing**.

What is the puzzle asking you to do?

Walk the path with your eyes from the red start block to the red end block. How many blocks should you place, and how many spots should you skip?

2 · The Big Idea: Place On / Place Off

The agent moves in a straight line, but should **not** place a block on every step. Placing is **on** at obsidian spots and **off** at glass spots.

So your code mixes `move forward` with `place block down`, and sometimes `move forward` alone.

```
place block down
move forward
place block down
move forward
move forward
place block down
```

Look at the code above. On which steps is placing ON? On which step is it OFF (a skipped spot)?

Why can't you just place a block on every single step?

3 · Plan Your Path

Plan on paper first. Mark each spot in the row **B** (build – place a block) or **S** (skip – move past, no block).

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Example: B B S B S S B

Write your row of B and S, in order from the red start to the red end:

Turn your plan into agent commands. A B spot needs a place and a move; an S spot needs only a move.

Write your commands:

- 1.
- 2.
- 3.
- 4.
- 5.
- 6.
- 7.
- 8.

4 • Spot the Bug 🐛

Each code block should copy the statue, but is broken. Rewrite it, then explain the fix.

Bug A – The pattern should be: build, skip, build. (B S B)

```
place block down
place block down
move forward
move forward
place block down
```

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B – The agent placed a block on a glass (skip) spot. The pattern should be: build, skip, skip, build. (B S S B)

```
place block down
move forward
place block down
move forward
place block down
move forward
place block down
```

Write the fixed code:

Why was it wrong? Why does your fix work?

5 • Show Your Work 📷🎥

Go to the Impossible Statue world and run your code. Match the crying obsidian exactly: place where it says build, leave the glass spots empty, and finish on the red end block.

Record **one video** (a phone is fine). Show two things:

1 • Your code. Scroll through it. Say what each part does.

2 • Your build. Point the camera. Name the parts.

Fill the blanks:

Today I built _____.

I built it using this code: _____.

In this code I used _____.

The hardest part was _____.

That part was hard because _____.

Write your lines here, then say them in your video.

Submit ☒

Send this worksheet + your video to teacher on KakaoTalk.

